Multiscale Difference Excitation Networks for Efficient Salient Object Detection in Remote Sensing Images

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***Abstract*—Developing lightweight architectures for optical remote sensing image salient object detection (ORSI-SOD) is essential to enable real-time interpretation of ORSI scenes on resource-constrained satellite platforms. Existing lightweight ORSI-SOD methods leverage lightweight encoders and carefully designed modules to enhance efficiency while maintaining performance. However, these methods often rely on intricate multistage feature aggregation strategies to mitigate background interference and enhance the feature response of salient objects, inevitably compromising the accuracy-efficiency balance. To address this issue, we propose the multiscale difference excitation network (MSDEN), a novel lightweight ORSI-SOD architecture that optimizes the accuracy-efficiency trade-off. Our approach first incorporates the multiscale difference excitation guidance module (MDEGM), which explicitly models local boundary cues and mitigates the suppression of object features caused by redundant backgrounds. Additionally, we employ the compact spatial feature fusion module (CSFM) to capture long-range contextual information, facilitating the refinement of salient object representations across diverse scales. We conduct extensive experiments on three public benchmark datasets, demonstrating that the proposed MSDEN substantially narrows the performance gap between lightweight models and recent state-of-the-art ORSI-SOD methods. It achieves the S_α of 0.9303 in the EORSSD dataset, with only 2.7M parameters and 1.0G FLOPs for 288 $\times$ 288 inputs.**

***Index Terms*—Boundary feature representation, lightweight salient object detection (SOD), optical remote-sensing image (OSRI).**

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I. Introduction

SALIENT object detection (SOD) aims to emulate the human visual system's capacity for rapid identification of visually distinctive objects/regions [1], [2], serving as a crucial preprocessing step for various computer vision tasks like object detection [3], [4], [5], [6], object tracking [7], [8], image quality assessment [9], [10], and image captioning [11], [12], which has been widely explored in various domains [10], [13], [14], [15], [16], [17]. For optical remote sensing images (ORSIs), SOD constitutes a fundamental preprocessing operation that notably improves large-scale scene interpretation and surface observation capabilities. It provides crucial support for remote sensing applications, such as disaster assessment, land monitoring, and urban planning [18]. This article focuses on developing SOD methods specifically for ORSIs (referred to as ORSI-SOD), addressing unique challenges in remote sensing scenarios.

Unlike natural scene images (NSIs), ORSIs encompass larger scenes due to elevated shooting angles. Objects within ORSIs display significant diversity in terms of types, scales, textures, and colors, posing significant challenges for directly adapting NSI-SOD methods to ORSI analysis. Recently, notable advancements have been made in the performance of ORSI-SOD methods. These methods typically employ convolutional neural networks (CNNs) or vision transformers (ViTs) with specialized feature enhancement modules to distinguish salient objects from complex backgrounds [19], [20], [21], [22], [23], [24]. However, the effectiveness of these methods often relies on heavy encoder parameters to extract salient object information from ORSIs, resulting in low computational speeds. For example, HFCNet [24] achieves state-of-the-art ORSI-SOD performance with 140.8M parameters and 120.4G FLOPs. Such excessive parameter counts and computational complexity pose significant challenges for deploying existing methods on resource-constrained satellite platforms. To address this issue, lightweight ORSI-SOD models have been developed to facilitate efficient deployment [22], [25], [26]. CorrNet [25] lightens the backbone (VGG-16) and builds a lightweight subnet for feature extraction. Although using only 4.09M parameters, it still runs with 21G FLOPs for computation. SeaNet [22], MEANet [26], and SAFINet [27] utilize MobileNetV2 as a lightweight encoder and design multiple strategies to aggregate multilevel features and enhance edge representation in spatial information. Despite the significant performance improvements demonstrated by these lightweight ORSI-SOD methods,

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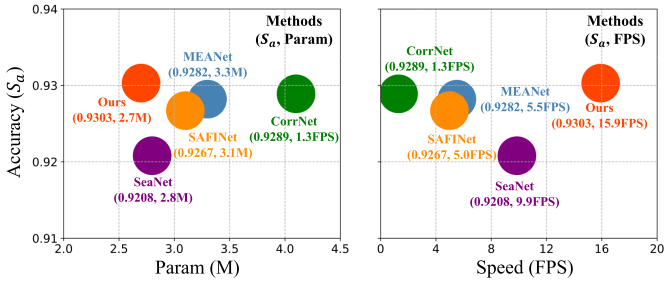


Fig. 1. Comparison of parameter size and inference speed [frames per second (FPS)] between our proposed method and existing state-of-the-art lightweight ORSI-SOD approaches. We convert the existing lightweight ORSI-SOD models into ONNX and deploy them on a HUAWEI Mate 70 device to evaluate their efficiency. The accuracy S_α is evaluated on the EORSSD dataset. The figure illustrates the competitive balance between accuracy and efficiency of our method.

several challenges remain to be addressed for real-world deployment.

A. Trade-Off Between Accuracy and Efficiency

Despite the recent advancements in lightweight ORSI-SOD approaches, such as MEANet and SAFINet, which have achieved significant reductions in parameters and FLOPs, their complex intermediate feature operations have notably compromised the inference speed (as illustrated in Fig. 1). In contrast, while SeaNet offers improved inference speed, it suffers from a substantial drop in accuracy. This highlights the critical challenge of developing efficient feature representation strategies that optimally balance computational efficiency with detection accuracy.

B. Complex Scene Context With Redundant Backgrounds

ORSIs inherently encompass expansive fields of view, which inevitably result in high background redundancy and sparse target distribution. Existing methods typically rely on large parameter backbones (e.g., VGGNet, swin transformer) to extract discriminative features from the extensive patterns present in ORSIs. However, lightweight models with limited capacity often struggle to effectively filter out redundant background features, notably hindering the accurate detection of salient objects.

C. Diverse Salient Objects With Weak Intrinsic Feature Responses

The salient features of objects in ORSIs, particularly edges and textures, are inherently susceptible to strong background suppression. This fundamental limitation leads to severely weak feature responses and poses a persistent challenge in detecting heterogeneous targets with varying appearances. Existing lightweight ORSI-SOD methods attempt to enhance target responses through sophisticated multilevel feature fusion strategies. However, these approaches exhibit limited efficacy in recovering suppressed structural and textural details. Consequently, this limitation restricts the representative capacity of extracted features, leading to a degradation in the detection performance.

To address these limitations, we introduce the multiscale difference excitation guidance module (MDEGM) to enhance the intrinsic feature response and mitigate background interference by refining boundary cues and incorporating deep feature guidance. Specifically, MDEGM first explicitly models the center-surround relationships within local regions to facilitate multiscale feature contrast measurement. Subsequently, it progressively mitigates background redundancy through high-level semantic guidance to highlight the boundary information of foreground objects. In addition to refining local regions, we design a compact spatial feature fusion module (CSFM) that captures long-range contextual dependencies through lightweight cross-scale feature interaction. CSFM notably improves the model's capacity for multiscale object recognition in ORSIs, effectively overcoming the inherent scale variations characteristic of ORSIs. Building on these designs, we propose multiscale difference excitation networks (MSDEN), an innovative lightweight architecture for ORSI-SOD that achieves a competitive balance between accuracy and efficiency compared to previous lightweight ORSI-SOD methods. As shown in Fig. 2(a), MSDEN comprises three components: 1) *Lightweight Encoder:* To reduce the number of parameters and computational cost of the framework, we adopt the MobileNet-V2 [28] as the lightweight backbone for efficiently capturing semantic information; 2) *Task-Specific Side Network:* The task-specific side network integrates the MDEGM and CSFM to enhance feature representation through local boundary information modeling and global feature refinement; and 3) *Lightweight Decoder:* The lightweight decoder progressively processes the semantic features of each stage to generate a final prediction. The effectiveness of MSDEN, as shown in Fig. 1, demonstrates a competitive efficiency-accuracy trade-off compared to the existing ORSI-SOD methods.

Our main contributions are as follows.

1) We propose an MDEGM designed to enhance the representation of high-order local contrast information in salient objects while mitigating interference from background redundancies in ORSIs.
2) With the proposed MDEGM, we design a lightweight ORSI-SOD architecture MSDEN. By effectively integrating local contrast with long-contextual information, MSDEN achieves a competitive balance between model accuracy and computational efficiency.
3) Extensive experiments on three benchmark datasets demonstrate that MSDEN attains superior performance while being more efficient than state-of-the-art methods. These results underscore the potential of our lightweight deep model for enabling efficient ORSI-SOD at the edge.

The remainder of this article is organized as follows. Section II reviews related work in both general ORSI-SOD and lightweight ORSI-SOD approaches. Section III presents our proposed methodology with detailed technical descriptions and analysis. The experimental validation, including comprehensive comparisons and ablation studies, is provided in Section IV. Section V discusses current limitations and potential future research directions. Finally, Section VI concludes the article.

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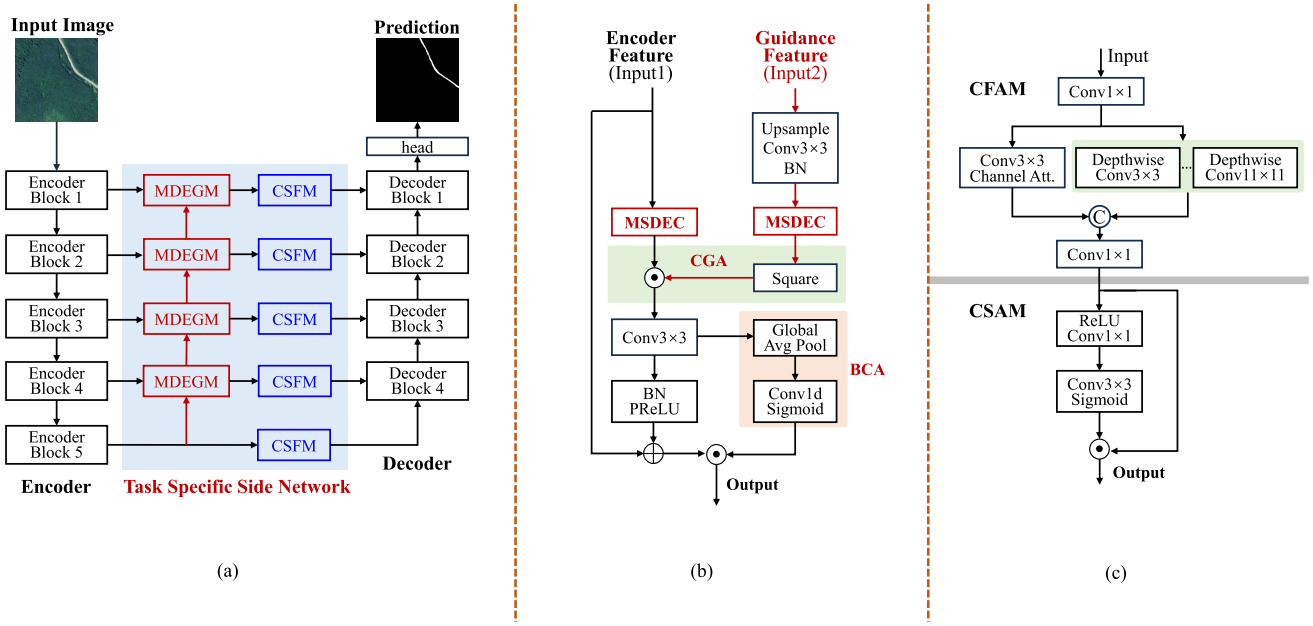


Fig. 2. Overall architecture of the proposed MSDEN. (a) Overall architecture of our proposed MSDEN consists of an MDEGM shown in (b) and a CSFM module shown in (c). (b) MDEGM consists of two MSDEC, a CGA module, and a BCA module. (c) CSFM consists of a CFAM and a CSAM.

II. Related Works

A. SOD in ORSIs

SOD serves as a critical preprocessing step for numerous visual tasks and has been extensively explored across various domains, including natural images [21], [29], [30], videos [13], [14], RGB-D data [15], [16], and ORSIs [18]. In the context of ORSIs, SOD functions as a tool for scene interpretation and surface observation, enabling support for a range of remote sensing applications, such as disaster monitoring, land use analysis, and urban development.

Compared to natural images, ORSIs present a wide variety of target types and scales within complex backgrounds, posing significant challenges for traditional SOD methods. To this end, numerous SOD approaches have been specifically developed for ORSI scenarios, incorporating carefully designed feature enhancement strategies. As a pioneering work, LVNet [19] proposed a two-stream pyramid module to perceive the diverse scales and local details in salient objects. PDFNet [20] explored multilevel feature interactions for ORSI-SOD. SARNet [31] roughly detected salient objects in the semantically guided decoder, followed by multilevel feature refinement to improve the final prediction accuracy. MJRBM [32] and EMFINet [21] designed edge-aware modules to preserve the boundary of salient objects from multilevel features. RRNet [33] designed a relational reasoning encoder to infer the semantic relationship in salient objects. ACCoNet [23] explored the coordination of adjacent features to capture contextual information. SFANet [34] adopted progressive semantic flow and uncertainty-aware refinement to obtain an accurate prediction. HFCNet [24] designed a new hybrid heterogeneous encoder that combined CNN and ViT to model local and global information.

Although the aforementioned methods have demonstrated excellent performance in the ORSI-SOD task, their large parameter sizes and computational complexity hinder their deployment on resource-constrained satellite platforms. Therefore, in this study, we aim to investigate a lightweight ORSI-SOD method to maintain competitive performance while substantially reducing computational requirements for practical applications.

B. Lightweight SOD in ORSIs

The lightweight SOD method was initially extensively explored in natural images. For example, CSNet [35] introduced a flexible convolutional module that hierarchically aggregated both in-stage and cross-stage multiscale features, achieving 80% parameter reduction while maintaining competitive accuracy. HVPNet [36] was inspired by the human vision system and proposed the hierarchical visual perception (HVP) module for better multiscale learning. SAMNet [37] adopted a stereoscopic attention mechanism to adaptively fuse multiscale features. Although these methods have achieved a fine balance between performance and efficiency in natural images, their lack of further enhancement and learning of intermediate features for diverse targets and complex scenes in ORSIs leads to significant performance degradation in ORSI-SOD tasks. To address this issue, CorrNet [25] lightened the VGG-16 for efficient feature extraction and adopted lightweight refinement modules to progressively generate saliency maps. Although CorrNet reduced the parameter count to 4.09M, it still required 21.1G FLOPs in computational cost. SeaNet [22], which adopted MobileNet-V2 as its feature encoder, introduces a dynamic semantic matching module and an edge self-alignment module to enhance feature representation. While SeaNet demonstrates improved computational efficiency, it exhibits notably inferior detection accuracy compared to state-of-the-art methods. To mitigate this limitation, MEANet [26] enhanced object representation by incorporating edge-aware features into a spatial attention mechanism. Similarly, SAFINet [27] employed an attention

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feedback module combined with a spatial correlation module to improve contextual feature aggregation. Although these lightweight architectures narrow the performance gap while preserving significant reductions in FLOPs and parameter counts, their reliance on intricate multilevel feature interaction results in suboptimal inference speeds.

Lightweight SOD for ORSIs presents unique challenges in balancing computational efficiency with detection accuracy due to complex background clutter and weak object feature response notably degrade model performance. To overcome this limitation, we propose a boundary-aware feature enhancement strategy that explicitly models salient object structures while suppressing background interference. Our approach enables resource-efficient ORSI-SOD architectures to achieve superior foreground–background discrimination without compromising computational performance.

C. Traditional Operators Enhance Convolution

Current CNN-based ORSI-SOD methods employ standard convolution operations to extract low-level visual features (e.g., textures and colors) and high-level semantic representations. These methods often fail to preserve crucial high-frequency boundary information. To this end, recent studies have developed gradient-enhanced convolutional variants inspired by local binary patterns (LBPs) [38], including central difference convolution (CDC) [39] and its advanced variant central pixel difference convolution (CPDC) [40], [41]. These convolutional kernels enable multidirectional gradient extraction. As demonstrated in previous works [39], [40], [41], these innovative operators notably outperform standard convolution in edge-aware feature representation while maintaining efficiency. The comparison of these gradient-enhanced convolutions and the normal convolution can be observed as follows:

$$y = f\left(\boldsymbol{x}, \boldsymbol{\theta}\right) = \sum_{x_i \in \mathcal{P}} w_i \cdot x_i, \quad \text{(norm convolution)} \tag{1}$$

$$y = f\left(\Delta\boldsymbol{x}, \boldsymbol{\theta}\right) = \sum_{x_i \in \mathcal{P}} w_i \cdot (x_i - \theta * x_0), \quad \text{(CDC)} \tag{2}$$

$$y = f\left(\Delta\boldsymbol{x}, \boldsymbol{\theta}\right) = \sum_{x_i \in \mathcal{P}} w_i \cdot (x_i - x_0), \quad \text{(CPDC)} \tag{3}$$

where $\boldsymbol{x}$ and $\boldsymbol{\theta}$ represent input in a local region $\mathcal{P}$ and convolutional kernel weights, respectively. x_i is the input pixels and x_0 is the central pixel in the local region $\mathcal{P}$, and w_i is the weight of a 3×3 convolutional kernel. Hyperparameter $\theta \in [0, 1]$ utilized in (2) trade-offs the contribution between the intensity-level and gradient-level information.

Based on these designed gradient-aware convolutions and diffusion theory, [42] and [43] build a semantic difference mechanism to improve the sensitivity to interclass boundaries rather than all the edges. However, these methods [42], [43] pose the challenge in the diversity scales of objects and complex scenes in ORSIs. Therefore, we introduce the multiscale semantic difference mechanism for ORSIs to enhance suppressed boundary cues caused by background interference and fully leverage high-order information of objects to improve feature responses of salient objects.

D. Artificial Intelligence in Remote Sensing Image Analysis

Remote sensing image analysis is critical for Earth's surface observation, with widespread applications in natural resource management, urban development, and disaster monitoring. Recent progress in artificial intelligence (AI) has notably improved the efficiency and accuracy of remote sensing image analysis. This section offers an overview of AI applications in remote sensing, covering optical, spectral, infrared, and multimodal images while highlighting recent developments and technical challenges in these areas.

For optical images, RingMo [44] utilized the MIM pretraining method to mitigate the challenge of detecting dense and small objects in complex remote sensing scenes, demonstrating competitive performance in various tasks. Spectral imaging can provide valuable analysis for the observed material composition. To extract knowledge representations from complex spatial–spectral data, SpectralGPT [45] was trained on the one million spectral images with more than 600 million parameters, facilitating the processing of large-scale spectral data and adapting to diverse inputs regarding size, resolution, temporal variations, and geographical coverage. To address the issues of weak texture and small object size in infrared images, UIUNet [46] embedded a tiny U-Net within a larger U-Net backbone, capturing multilevel target features. To improve the generalizability of remote sensing models in multimodal data, SkySense [47] enhanced cross-modal generalization through a novel spatiotemporal encoder that jointly processed optical-SAR time series via multigranularity contrastive learning, effectively aligning feature representations across modalities and spatial scales. In general, remote sensing foundation models employ self-supervised learning paradigms such as masked reconstruction or contrastive learning to derive generalizable representations from large-scale remote sensing data, capturing intrinsic feature correlations and enhancing downstream task performance. Specifically, alignment learning for multimodal inputs further strengthens cross-modal representations. Moreover, task-oriented models focus on multiscale learning and global–local interactions to improve recognition performance within specific modalities and tasks.

Despite their distinct challenges, these tasks collectively underscore the critical need to develop universal feature representations with robust cross-domain generalization capabilities in remote sensing. Future research in remote sensing should prioritize innovative areas such as cross-modal and multitask learning, facilitating adaptation to downstream tasks in complex scenarios.

III. Methods

In this section, we present a detailed description of our proposed lightweight MSDEN. We begin by providing an overview of the architecture in Sec. III-A. Subsequently, in Sec. III-B and Sec. III-C, we present detailed information on designed feature enhancement modules: the MDEGM and the CSFM. Furthermore, in Sec. III-D, we explain the decoder structure and the training loss function in our approach.

A. Overall Architecture of MSDEN

As shown in Fig. 2(a), the proposed MSDEN is based on a general bottom-up pathway architecture and consists of three

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components: a lightweight feature encoder, a task-specific side network that includes the MDEGM and the CSFM, and a lightweight decoder. The task-specific side network first refines the intrinsic feature representation of salient objects by explicitly modeling boundary cues and subsequently captures long-contextual spatial information to mitigate the impact of scale diversity.

Specifically, we utilize the MobileNet as a lightweight encoder following [22], [26], [48]. The encoder is divided into five stages, denoted as $E^i, (i = 1, 2, 3, 4, 5)$. The output features of each stage are denoted as $\mathcal{F}^i \in \mathbb{R}^{C_i \times H_i \times W_i}$. In the ith stage, the encoder feature $\mathcal{F}^i$ is processed by a task-specific side network before being fed into the decoder. We denote the MDEGM's output at stage i as $\mathcal{G}^i$. In stage i, the MDEGM performs a feature interaction between the encoder feature $\mathcal{F}^i$ and the guidance feature $\mathcal{G}^{i+1}$ from stage $i+1$. In particular, $\mathcal{F}^5$ is directly used as the guidance feature in stage 4. The MDEGM first refines local contrast representations in both $\mathcal{F}^i$ and guidance feature $\mathcal{G}^{i+1}$, then enhances intrinsic object features while suppressing redundant information in $\mathcal{F}^i$ through deep guidance feature modulation. MDEGM focuses on modeling local features and overlooks the capture of long-range features crucial for larger objects, such as bridges. To address this limitation, we introduce the CSFM, which efficiently captures long-range contextual information. The decoder of our MSDEN consists of four lightweight blocks, which are denoted as D^i. Each D^i employs 1×1 convolution block to aggregate information from neighboring task-specific side networks. Finally, a head module is employed at the end of the decoder blocks to generate the predicted saliency map.

B. Multiscale Difference Excitation Guidance Module

Based on the aforementioned limitation, lightweight ORSI-SOD methods pose challenges in capturing discriminative features, as intrinsic object feature responses (such as boundary cues) are often suppressed and interfered with by complex backgrounds. Therefore, we refine the boundary cues of the salient objects to improve the structural response and mitigate the impact of redundancy. To achieve this goal, we propose the MDEGM. The architecture can be observed in Fig. 2(b). The MDEGM consists of three key components: 1) multiscale difference excitation convolution (MSDEC) to generate rich features with high-order information by leveraging multiscale pixel contrast cues instead of pixel intensity cues as in standard convolution; 2) cross gate attention (CGA) to suppress the impact of redundant background features by deep semantic feature guidance; and 3) boundary-aware channel attention (BCA) to strengthen the activation of feature channels that convey more boundary-related information of objects. The process of MDEGM in the i-stage can be formulated as

$$\mathcal{G}^i = \mathrm{MDEGM}_i\left(\mathcal{F}^i, \mathcal{G}^{i+1}\right) \tag{4}$$

where the $\mathcal{G}^i$ is the output feature of MDEGM in the i-stage. Specifically, MDEGM at stage i leverages the MSDEC component to hierarchically extract multiscale local contrast features. This operation processes two distinct feature inputs: the encoder output features $\mathcal{F}^i$ from the current stage and the deep guidance features $\mathcal{G}^{i+1}$ generated by MDEGM at the subsequent stage $i+1$. In particular, in stage 4, the encoder output feature $\mathcal{F}^5$ is directly utilized as the guidance feature in MDEGM. Subsequently, CGA is employed to enhance salient object representations in the current layer by adaptively weighting spatial features using the refined guidance features generated through MSDEC. Finally, BCA performs channel-wise recalibration to amplify boundary-sensitive object representations. The details of each component in MDEGM are presented as follows.

1) Multiscale Difference Excitation Convolution: Building upon the achievements of convolution operators incorporating traditional image gradient operators [41], [43], we design the MSDEC to extract local contrast information, addressing the scale diversity of objects in ORSIs. To reduce computational complexity, MSDEC adopts the depthwise convolution operation. The comparison with the standard depthwise convolution is shown in Fig. 3. MSDEC comprises multiple groups of different excitation convolution (DEC) with varying dilation rates. Compared to the standard depthwise convolution filter, the DEC employs a kernel with -1 as the fixed center value to extract local feature contrast cues from objects. This DEC can be viewed as a dynamic weighted fusion of neighboring pixels, followed by a differencing operation with the central pixel, which effectively preserves both intensity-level and gradient-level feature information, including structural details. Based on this design, MSDEC enhances the sensitivity of convolution to pixel differences within local regions while accommodating the scale diversity of objects in ORSIs. The further analysis is provided in Sec. IV-D.

Given the input feature $\mathcal{F} \in \mathbb{R}^{C \times H \times W}$, we first split $\mathcal{F}$ into N groups $[\mathcal{F}_1, \ldots, \mathcal{F}_k, \ldots, \mathcal{F}_N]$ along the channel dimension, where each group $\mathcal{F}_k \in \mathbb{R}^{C/N \times H \times W}$ contains C/N channels. Subsequently, each group of features is associated with a set of DECs where dilation rates range from 1 to N. The process of MSDEC$(\cdot)$ can be formulated as follows:

$$\begin{aligned} \{\mathcal{F}_k\} &= \mathrm{Split}\left[\mathcal{F}\right], k = 1 : N \\ \mathcal{F}_k^{out} &= \mathrm{DEC}\left(\mathcal{F}_k; \theta; d = k\right) \\ \mathcal{F}^{out} &= \mathrm{Cat}\left[\mathcal{F}_1^{out}, \ldots, \mathcal{F}_k^{out}, \ldots, \mathcal{F}_N^{out}\right] \end{aligned} \tag{5}$$

where Split$[\cdot]$ is the channel split operation, $\mathcal{F}^{out}$ is the output of MSDEC, $\theta \in \mathbb{R}^{C/N \times 1 \times 3 \times 3}$ is the weight of DEC, d is the dilation of the DEC in kth group, and Cat$[\cdot]$ denotes the concatenation in channel dimension.

In MDEGM of i-stage, we first refine the local contrast cues of encoder feature $\mathcal{F}^i$ and deep guidance feature $\mathcal{G}^{i+1}$ by MSDEC, which can be formulated as follows:

$$\begin{aligned} \mathcal{F}_{edge}^i &= \mathrm{MSDEC}\left(\mathcal{F}^i\right) \\ \mathcal{G}_{edge}^{i+1} &= \mathrm{MSDEC}\left(\mathrm{BN}\left(\mathrm{Conv}_{3\times3}\left(\mathrm{Up}\left(\mathcal{G}^{i+1}\right)\right)\right)\right) \end{aligned} \tag{6}$$

where Up$(\cdot)$ denotes the upsample operation, $\mathrm{Conv}_{3\times3}$ denotes the 3×3 Convolution. The combination of the upsampling layer followed by a 3×3 convolution is to align spatial and channel dimensions of $\mathcal{G}^{i+1}$ with those of $\mathcal{F}^i$.

2) Cross Gate Attention: After passing through MSDEC, $\mathcal{F}_{edge}^i$ contains rich boundary-enhanced features of objects within the scene. However, the representation of salient objects remains constrained due to interference from redundant background information. To address this, MDEGM leverages CGA

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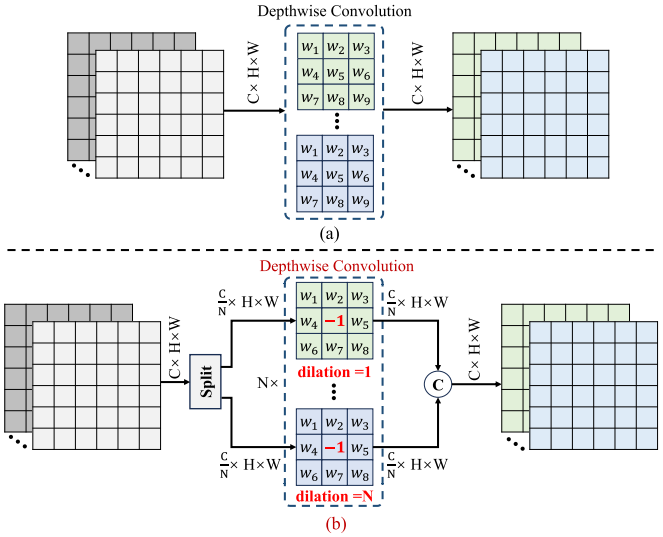


Fig. 3. (a) Depthwise convolution with standard convolutional filters. (b) Details of MSDEC. MSDEC first divides input features into N channel groups. Each group undergoes a depthwise convolution with a center of -1 at a specific dilation rate, after which all groups are concatenated to form the output.

to effectively filter out redundant background information by refined deep guidance features $\mathcal{G}_{edge}^{i+1}$, thus enhancing the boundary cues of salient objects. Specifically, the square term of $\mathcal{G}_{edge}^{i+1}$ is used to refine the structural response of salient objects and mitigate the interference of redundant background information in $\mathcal{F}_{edge}^{i}$. The CGA process can be formulated as

$$\mathcal{G}_{s}^{i} = \mathrm{Conv}_{3\times 3}\left(\mathcal{F}_{edge}^{i} \odot \mathcal{G}_{edge}^{i+1} \odot \mathcal{G}_{edge}^{i+1}\right) \tag{7}$$

where $\odot$ denotes the element-wise multiplication.

3) Boundary-Aware Channel Attention: The enhanced feature $\mathcal{G}_{s}^{i}$, which incorporates guided boundary structure information of the salient objects, is combined with the original feature $\mathcal{F}^{i}$ to highlight the salient regions. To further emphasize channels that contain critical target information, we leverage these boundary-rich features to generate adaptive coefficients that selectively amplify feature channels with stronger boundary responses. This BCA mechanism builds on the attention function introduced in [49], and can be formally expressed as

$$\begin{aligned} s &= \mathrm{Sigmoid}\left(\mathrm{Conv1D}_{k}\left(\mathrm{GAP}\left(\mathcal{G}_{s}^{i}\right)\right)\right) \\ \mathcal{G}^{i} &= s \cdot \left(\mathrm{PReLU}\left(\mathrm{BN}\left(\mathcal{G}_{s}^{i}\right) + \mathcal{F}^{i}\right)\right) \end{aligned} \tag{8}$$

where $s \in \mathbb{R}^{C_i\times 1\times 1}$ is the channel-wise scale factor, $\mathcal{G}^{i} \in \mathbb{R}^{C_i\times H_i\times W_i}$ is the final output of MDEGM in the stage i, $\mathcal{F}^{i}$ is the residual term, GAP$(\cdot)$ is the Global Average Pooling layer, and Conv1D$_k$ denotes the 1-D convolution with kernel size k. In this article, the k is empirically set to 3.

In MDEGM, semantic difference features are propagated to boundary regions through multiscale difference feature extraction and hierarchical feature interaction. Specifically, we first employ MSDEC (with a center kernel weight of -1) to encode the differential relationship between the central feature value and its multiscale neighboring feature values, highlighting boundary responses of local features and mitigating interference from redundant background information.

Aspects	Stage1	Stage2	Stage3	Stage4	Stage5
Input size	[16 × 144 × 144]	[24 × 72 × 72]	[32 × 36 × 36]	[96 × 18 × 18]	[320 × 9 × 9]
Conv1 × 1	(16, 12)	(24, 12)	(32, 12)	(96, 12)	(320, 12)
DepthwiseConv	(12, 3, 3) (12, 5, 5) (12, 7, 7)	(12, 3, 3) (12, 5, 5) (12, 7, 7) (12, 9, 9)	(12, 3, 3) (12, 5, 5) (12, 7, 7) (12, 9, 9) (12, 11, 11)	(12, 3, 3) (12, 5, 5) (12, 7, 7) (12, 9, 9) (12, 11, 11)	(12, 3, 3) (12, 5, 5) (12, 7, 7)
Output size	[24 × 144 × 144]	[24 × 72 × 72]	[24 × 36 × 36]	[24 × 18 × 18]	[24 × 9 × 9]

Fig. 4. Configuration of CFAM in MSDEN. The (d_1,12) in the row of conv1 × 1 denotes the input channel d_1 and output channel 12. (12, k, k) in the row of depthwiseconv denotes the depthwise convolution $W \in \mathbb{R}^{12\times 1\times k\times k}$ used in CFAM. The k is the kernel size of depthwise convolution.

Furthermore, as deeper features tend to focus more on target regions, we leverage deep features as guidance signals and enhance boundary responses of object regions in shallow features through adjacent-layer feature interactions. The MDEGM effectively improves the ORSI-SOD performance of the lightweight model.

C. Compact Spatial Feature Fusion Module

The MDEGM effectively enhances local contrast information extraction for the salient object in ORSIs. However, it still lacks sufficient interaction with pixels within a large receptive field to effectively accommodate large salient objects, such as bridges. Therefore, we integrate a CSFM following MDEGM to efficiently capture long-range contextual information.

As shown in Fig. 2, CSFM consists of two parts: compact feature aggregation module (CFAM) and compact spatial attention module (CSAM). To reduce computational cost, the CFAM first projects the output features $\mathcal{G}^{i} \in \mathbb{R}^{C_i\times H_i\times W_i}$ of MDEGM in the stage i into a low channel dimension (i.e., 12) by a 1 × 1 convolution, which can be formulated as

$$\mathcal{G}_{proj}^{i} = \mathrm{Conv}_{1\times 1}\left(\mathcal{G}^{i}\right) \tag{9}$$

where Conv$_{1\times1}$ denotes one 1 × 1 convolution layer, and $\mathcal{G}_{proj}^{i} \in \mathbb{R}^{12\times H_i\times W_i}$ in this article.

Then, the feature is processed by two branches. One branch is utilized to enhance the channel representation, and another branch includes a group of depthwise convolution layers with various kernel sizes to capture and aggregate multiscale spatial features. The process can be formulated as

$$\begin{aligned} \mathcal{G}_{c}^{i} &= \mathrm{CA}\left(\mathrm{Conv}_{3\times 3}\left(\mathcal{G}_{proj}^{i}\right)\right) \\ \mathcal{G}_{m}^{i} &= \sum_{k\in\mathcal{K}_i} \mathrm{DwConv}_{k\times k}\left(\mathcal{G}_{proj}^{i}\right) \\ \mathcal{G}_{CFAM}^{i} &= \mathrm{Conv}_{1\times 1}\left(\mathrm{Cat}\left[\mathcal{G}_{m}^{i}, \mathcal{G}_{c}^{i}\right]\right) \end{aligned}$$

where $\mathcal{G}_{CFAM}^{i} \in \mathbb{R}^{24\times H_i\times W_i}$, CA$(\cdot)$ denotes the channel attention in [49], Cat$[\cdot]$ denotes the concatenation in channel dimension, DwConv$_{k\times k}(\cdot)$ is the depthwise convolution with kernel size k, and the $\mathcal{K}_i$ is the set of depthwise convolutional kernels used by CFAM in the ith stage. The details of $\mathcal{K}$ can be seen in Fig. 4.

After CFAM, CSAM is utilized to improve the feature response of salient objects and filter feature noise by spatial attention. The final output of CSFM can be formulated as

$$\begin{aligned} \alpha^{i} &= \mathrm{Sigmoid}\left(\mathrm{Conv}_{3\times 3}\left(\mathrm{Conv}_{1\times 1}\left(\mathrm{ReLU}\left(\mathcal{G}_{CFAM}^{i}\right)\right)\right)\right) \\ \mathcal{F}_{t}^{i} &= \alpha^{i} \cdot \mathcal{G}_{CFAM}^{i} \end{aligned} \tag{10}$$

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where $\text{Conv}_{1\times1}$ reduces the channel of input to 4, and then $\text{Conv}_{3\times3}$ followed the Sigmoid function generates the spatial attention map $\alpha^i \in \mathbb{R}^{1\times H_i\times W_i}$, $\mathcal{F}_t^i$ is the final output of task-specific side network in i-stage.

The advantages of CSFM are: 1) building upon the saliency-aware features generated by MDEGM, the CSFM captures both long- and short-range contextual dependencies, thereby enhancing its adaptability to scale variations of objects in ORSIs, and 2) the CSFM is computationally efficient, requiring only 0.03M parameters and achieving a 0.10G FLOPs for 288 × 288 input resolution.

D. Decoder and Loss Function

1) Decoder: Based on the highlighted salient object areas provided by the previous two modules, we design a lightweight decoder to progressively generate saliency maps. As shown in Fig. 2, our decoder consists of four blocks (i.e., D^1, D^2, D^3, and D^4, and a head module). Each decoder block contains one 1 × 1 convolution block and an upsampling operation. In contrast to the 3 × 3 convolution or 3 × 3 depthwise convolution employed in previous methods, we have substituted the convolutions in the decoding layers with 1 × 1 convolutions. This substitution effectively reduces the computational cost and parameter size in the decoding part, facilitating the feature representative capability by expanding the capacity of the task-specific side network. The decoder process in the stage i can be formulated as

$$\mathcal{F}_d^i = \text{PReLU}\left(\text{BN}\left(\text{Conv}_{1\times1}\left(\text{Cat}\left[\mathcal{F}_t^i, \text{Up}\left(\mathcal{F}_d^{i+1}\right)\right]\right)\right)\right) \quad (11)$$

where $\mathcal{F}_d^i$ denotes the output of decoder in ith stage, Cat[·] denotes the concatenation in channel-dimension, and Up(·) is the bilinear interpolation to align the spatial dimensions of $\mathcal{F}_d^{i+1}$ and $\mathcal{F}_t^i$. Finally, the *head* module generates the final predicted saliency map. The head module is composed of a dropout layer and a 1 × 1 convolution layer, which can be formulated as follows:

$$\mathcal{F}_{pred} = \text{Conv}_{1\times1}\left(\text{Dropout}\left(\text{Up}\left(\mathcal{F}_d^1\right)\right)\right). \quad (12)$$

2) Loss Function: Following previous methods [22], [25], we adopt the binary cross-entropy (BCE) loss and intersection-over-union (IoU) loss to jointly train our MSDEN. The total loss can be expressed as

$$L_{total} = L_{bce}\left(\mathcal{F}_{pred}, Y_{true}\right) + L_{iou}\left(\mathcal{F}_{pred}, Y_{true}\right) \quad (13)$$

where L_{bce} and L_{iou} are BCE loss and IoU loss, respectively, $\mathcal{F}_{pred}$ is the final output of model, and Y_{true} is the ground truth (GT).

IV. Experiments

A. Experiment Protocol

1) Datasets: In this article, we conduct experiments on ORSSD [19], EORSSD [50], and ORSI-4199 [32] datasets. The ORSSD dataset is the first dataset specifically for the ORSI-SOD task, which has 600 images for training and 200 images for testing. The EORSSD dataset is an extended version of ORSSD, which adds 1200 ORSIs to the ORSSD dataset, resulting in 2000 ORSIs. This dataset contains 1400 images for training and 600 images for testing. ORSI-4199 contains 2000 training images and 2199 testing images. All three datasets utilize pixel-level annotations to identify salient objects for each ORSI.

2) Evaluation Metrics: Following previous methods [22], [25], we compare the proposed method with previous methods in terms of quantitative evaluation metrics and computational complexity. For quantitative evaluation metrics, S-measure ($S_\alpha, \alpha = 0.5$) [51], F-measure ($F_\beta, \beta^2 = 0.3$) [52], E-measure (E_ξ) [53], and mean absolute error ($\mathcal{M}$). The S_α evaluates the similarity of regions of objects in parts and entire objects. F_β is calculated as the weighted average of precision and recall. E_ξ measures the similarity between the predictions and the labels by considering local and image-level saliency values simultaneously. $\mathcal{M}$ measures the average absolute difference at all pixel locations. Computational complexity metrics include the inference speed measured in FPS, the parameter amount (Param) measured in millions (M), and the number of FLOPs measured in giga (G). The inference speed is reported with a batch size of 1 and no I/O time.

3) Implement Details: We conducted the experiments based on Pytorch and trained/tested the proposed method on a server with Xeon Gold 6230 CPU and NVIDIA V100 GPU. The network is optimized by Adam. The batch size is set to 8, and the initial learning rate is 1e-4. The total training epoch is 60. The learning rate is reduced by a factor of 10 after 30 epochs for EORSSD/ORSI-4199 and after 40 epochs for ORSSD. We set the input size to 288 × 288 following SeaNet [22]. The data augmentation strategy includes flipping and rotation following [22]. Consistent with previous lightweight ORSI-SOD approaches [22], [26], [27], we default to adopting MobileNet-V2 as our encoder, initialized with pretrained parameters. To ensure a fair comparison with [48], we also evaluate the performance of our MSDEN when using the MobileNet-V3 encoder. The standard practice of training and testing in separate sets for each dataset is employed as described in [22].

B. Evaluation Results on the ORSI-SOD Task

We compare our MSDEN with 23 state-of-the-art conventional SOD methods, including 10 conventional NSI-SOD methods (DSS [29], RADF [54], R3Net [55], PoolNet [56], GCPA [57], MINet [58], ITSD [59], GateNet [60], SUCA [61], and PA-KRN [62]), and 13 conventional ORSI-SOD methods (LVNet [19], DAFNet [50], SARNet [31], MJRBM [32], EMFINet [21], ACCoNet [23], ERPNet [63], SFANet [34], HFCNet [24], LGIPNet [64], MRBINet [65], BCARNet [66], and MambaDif [67]). Furthermore, the 11 state-of-the-art lightweight SOD methods, including four lightweight NSI-SOD methods (CSNet [35], SAMNet [37], HVPNet [36], and ADMNet [68]) and seven lightweight ORSI-SOD methods (CorrNet [25], SeaNet [22], MEANet [26], SAFINet [27], ISAANet [48], SOLNet [69], and LiteSalNet [70]), are selected to compare with our MSDEN. The above NSI-SOD methods are trained on the ORSI-SOD datasets following the training settings in [22].

1) Compared With Conventional SOD Methods: The performance comparison with conventional SOD methods can be found at the top of Tables I and II. Our MSDEN achieves the best performance in terms of accuracy and efficiency. For example, our method achieves 0.9303 and 0.9389 of S_α

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TABLE I

QUANTITATIVE EVALUATION. THE EVALUATION METRICS INCLUDE THE FOLLOWING. STRUCTURE-MEASURE (S_m), MAXIMUM, MEAN, AND ADAPTIVE F-MEASURE (F_β^{max}, F_β^{mean}, F_β^{ada}); MAXIMUM, MEAN, AND ADAPTIVE E-MEASURE (E_ξ^{max}, E_ξ^{mean}, E_ξ^{ada}); AND MAE ($\mathcal{M}$). FOR MAE SMALLER IS BETTER, WHILE FOR THE OTHER METRICS LARGER IS BETTER. THE BEST THREE RESULTS ARE INDICATED IN RED, BLUE, AND GREEN, RESPECTIVELY

Methods	Type	Backbone	Params (M) ↓	FLOPs (G) ↓	EORSSD								ORSSD							
					$S_\alpha\uparrow$	$F_\beta^{max}\uparrow$	$F_\beta^{mean}\uparrow$	$F_\beta^{adp}\uparrow$	$E_\xi^{max}\uparrow$	$E_\xi^{mean}\uparrow$	$E_\xi^{adp}\uparrow$	$\mathcal{M}\downarrow$	$S_\alpha\uparrow$	$F_\beta^{max}\uparrow$	$F_\beta^{mean}\uparrow$	$F_\beta^{adp}\uparrow$	$E_\xi^{max}\uparrow$	$E_\xi^{mean}\uparrow$	$E_\xi^{adp}\uparrow$	$\mathcal{M}\downarrow$
DSS_{17} [29]	CN	VGG16	62.2	114.6	.7868	.6849	.5801	.4597	.9186	.7631	.6933	.0186	.8262	.7467	.6962	.6206	.8860	.8362	.8085	.0206
$RADF_{18}$ [54]	CN	VGG16	62.5	214.2	.8179	.7446	.6582	.4933	.9140	.8567	.7162	.0168	.8259	.7619	.6856	.5730	.9130	.8298	.7678	.0382
$R3Net_{18}$ [55]	CN	ResNeXt	56.2	47.5	.8184	.7498	.6302	.4165	.9483	.8294	.6462	.0171	.8141	.7456	.7383	.7379	.8913	.8681	.8887	.0399
$PoolNet_{19}$ [56]	CN	VGG16	53.6	123.4	.8207	.7545	.6406	.4611	.9292	.8193	.6836	.0210	.8403	.7706	.6999	.6166	.9343	.8650	.8124	.0358
$GCPA_{20}$ [57]	CN	ResNet50	67.1	54.3	.8869	.8347	.7905	.6723	.9524	.9167	.8647	.0102	.9026	.8687	.8433	.7861	.9509	.9341	.9205	.0168
$MINet_{20}$ [58]	CN	VGG16	47.6	146.3	.9040	.8344	.8174	.7705	.9442	.9346	.9243	.0093	.9040	.8761	.8574	.8251	.9545	.9454	.9423	.0144
$ITSD_{20}$ [59]	CN	VGG16	17.1	54.5	.9050	.8523	.8221	.7421	.9556	.9407	.9103	.0106	.9050	.8735	.8502	.8068	.9601	.9482	.9335	.0165
$GateNet_{20}$ [60]	CN	ResNet50	100.0	108.3	.9114	.8566	.8228	.7109	.9610	.9385	.8909	.0095	.9186	.8871	.8679	.8229	.9664	.9538	.9428	.0137
$SUCA_{21}$ [61]	CN	ResNet50	117.7	56.4	.8988	.8229	.7949	.7260	.9520	.9277	.9082	.0097	.8989	.8484	.8237	.7748	.9584	.9400	.9194	.0145
$PA\text{-}KRN_{21}$ [62]	CN	ResNet50	141.1	617.7	.9192	.8639	.8358	.7993	.9616	.9536	.9416	.0104	.9239	.8890	.8727	.8548	.9680	.9620	.9579	.0139
$LVNet_{19}$ [19]	CR	VGG16	-	-	.8630	.7794	.7328	.6284	.9254	.8801	.8445	.0146	.8815	.8263	.7995	.7506	.9456	.9259	.9195	.0207
$DAFNet_{21}$ [50]	CR	VGG16	29.4	68.5	.9166	.8614	.7845	.6427	.9861	.9291	.8446	.0060	.9191	.8928	.8511	.7876	.9771	.9539	.9360	.0113
$SARNet_{21}$ [31]	CR	VGG16	25.9	129.7	.9240	.8719	.8541	.8304	.9620	.9555	.9536	.0099	.9134	.8850	.8619	.8512	.9557	.9477	.9464	.0187
$MJRBM_{22}$ [32]	CR	VGG16	43.5	95.7	.9197	.8656	.8239	.7066	.9646	.9350	.8897	.0099	.9204	.8842	.8566	.8022	.9623	.9415	.9328	.0163
$EMFINet_{22}$ [21]	CR	VGG16	107.3	480.9	.9290	.8720	.8486	.7984	.9711	.9604	.9501	.0084	.9366	.9002	.8856	.8617	.9737	.9671	.9663	.0109
$ACCoNet_{23}$ [23]	CR	VGG16	127.0	51.3	.9290	.8837	.8552	.7969	.9727	.9653	.9450	.0074	.9437	.9149	.8971	.8806	.9796	.9754	.9721	.0080
$ERPNet_{23}$ [63]	CR	VGG16	77.2	171.9	.9252	.8743	.8269	.7170	.9665	.9366	.8994	.0082	.9352	.9036	.8798	.8392	.9738	.9604	.9531	.0114
$SFANet_{24}$ [34]	CR	Res2Net	25.1	7.7	.9349	.8833	.8680	.8492	.9769	.9726	.9681	.0058	.9453	.9192	.9063	.8984	.9830	.9789	.9799	.0077
$HFCNet_{24}$ [24]	CR	Hybrid	140.8	120.4	.9449	.8951	.8646	.8051	.9842	.9719	.9386	.0049	.9543	.9269	.9049	.8710	.9884	.9798	.9748	.0070
$LGIPNet_{25}$ [64]	CR	PVT	52.1	47.0	.9392	.8924	.8770	.8623	.9806	.9753	.9719	.0053	.9507	.9266	.9129	.9027	.9876	.9829	.9809	.0066
$MRBINet_{25}$ [65]	CR	Res2Net	32.4	42.8	.9351	.8852	.8768	.8626	.9766	.9721	.9729	.0056	.9474	.9199	.9110	.9058	.9851	.9812	.9793	.0069
$BCARNet_{25}$ [66]	CR	ResNet	24	7	.9360	.8871	.8736	.8740	.9761	.9728	.9734	.0056	.9466	.9196	.9070	.9073	.9833	.9799	.9796	.0071
$MambaDif_{26}$ [67]	CR	VMamba	38.0	4.2×10^3	.9450	.9002	.8770	.8719	.9839	.9776	.9774	.0039	.9544	.9300	.9161	.9148	.9876	.9838	.9827	.0056
$CSNet_{20}$ [35]	LN	-	0.1	0.7	.8364	.8341	.7656	.6319	.9535	.8929	.8339	.0169	.8910	.8790	.8285	.7615	.9628	.9179	.9068	.0186
$SAMNet_{21}$ [37]	LN	-	1.3	0.5	.8622	.7813	.7214	.6114	.9421	.8700	.8284	.0132	.8761	.8137	.7531	.6843	.9478	.8818	.8656	.0217
$HVPNet_{21}$ [36]	LN	-	1.23	1.1	.8734	.8036	.7377	.6202	.9482	.8721	.8270	.0110	.8610	.7938	.7396	.6726	.9320	.8717	.8471	.0225
$ADMNet_{24}$ [68]	LN	-	0.8	3.3	.8784	.7761	.7670	.7473	.9194	.9010	.9091	.0099	.9060	.8487	.8392	.8276	.9480	.9360	.9430	.0168
$CorrNet_{22}$ [25]	LR	Lighten VGG	4.1	21.1	.9289	.8778	.8620	.8311	.9696	.9646	.9593	.0083	.9380	.9129	.9002	.8875	.9790	.9746	.9721	.0098
$SeaNet_{23}$ [22]	LR	MobileNetV2	2.8	1.7	.9208	.8649	.8519	.8304	.9710	.9651	.9602	.0073	.9260	.8942	.8772	.8625	.9767	.9722	.9670	.0105
$MEANet_{24}$ [26]	LR	MobileNetV2	3.3	9.6	.9282	.8766	.8678	.8557	.9708	.9658	.9663	.0070	.9340	.9033	.8934	.8860	.9768	.9730	.9715	.0098
$SAFINet_{24}$ [27]	LR	MobileNetV2	3.1	7.6	.9267	.8799	.8710	.8575	.9732	.9682	.9682	.0065	.9401	.9106	.9030	.9000	.9786	.9748	.9737	.0086
$ISAANet_{24}$ [48]	LR	MobileNetV3	3.6	4.3	.9351	.8869	.8748	.8643	.9761	.9717	.9734	.0063	.9429	.9166	.9034	.8982	.9836	.9787	.9777	.0077
$SOLNet_{25}$ [69]	LR	RepVGG	6.5	8.1	.9171	.8609	.8513	.8392	.9623	.9577	.9596	.0078	.9284	.9012	.8946	.8925	.9734	.9689	.9697	.0111
$LiteSalNet_{25}$ [70]	LR	MobileNetV2	3.9	7.4	.9273	.8877	.8809	.8796	.9743	.9634	.9729	.0063	.9372	.9124	.9061	.9056	.9789	.9700	.9752	.0090
MSDEN$_{288\times288}$	LR	MobileNetV2	2.7	1.0	.9303	.8817	.8674	.8456	.9780	.9729	.9709	.0069	.9389	.9076	.8920	.8786	.9793	.9749	.9705	.0089
† **MSDEN**$_{288\times288}$	LR	MobileNetV2	2.7	1.0	.9341	.8870	.8732	.8520	.9768	.9722	.9711	.0063	.9400	.9124	.8980	.8857	.9775	.9733	.9714	.0091
† **MSDEN**$_{288\times288}$	LR	MobileNetV3	4.2	1.1	.9359	.8875	.8716	.8485	.9786	.9742	.9692	.0057	.9445	.9120	.8989	.8887	.9805	.9764	.9749	.0094
† **MSDEN**$_{352\times352}$	LR	MobileNetV3	4.2	1.6	.9379	.8910	.8763	.8546	.9804	.9758	.9716	.0058	.9443	.9201	.9034	.8913	.9837	.9794	.9759	.0080

∗ **CN:** Conventional NSI-SOD method; **LN:** Lightweight NSI-SOD method; **CR:** Conventional ORSI-SOD method; **LR:** Lightweight ORSI-SOD method.
∗ $\text{MSDEN}_{h\times w}$ denotes the that the input image size is $h\times w$.
† Adding extra multi-scale training strategy employed in MEANet [26], SAFINet [27], ISAANet [48] and LiteSalNet [70].

TABLE II

QUANTITATIVE EVALUATION ON ORSI-4199. THE BEST THREE RESULTS ARE MARKED IN RED, BLUE, AND GREEN, RESPECTIVELY

Methods	Backbone	$S_\alpha\uparrow$	$\mathcal{M}\downarrow$	$F_\beta^{max}\uparrow$	$F_\beta^{mean}\uparrow$	$E_\xi^{max}\uparrow$	$E_\xi^{mean}\uparrow$
$PA\text{-}KRN_{21}$ [62]	ResNet50	.8428	.0385	.8423	.8257	.9241	.9121
$DAFNet_{21}$ [50]	VGG16	.8653	.0344	.8470	.8244	.9365	.9167
$MJRBM_{22}$ [32]	VGG16	.8582	.0372	.8511	.8305	.9343	.9071
$CorrNet_{22}$ [25]	Lighten VGG	.8623	.0366	.8560	.8513	.9330	.9206
$SeaNet_{23}$ [22]	MobileNetV2	.8673	.0348	.8643	.8591	.9423	.9363
$SFANet_{24}$ [34]	Res2Net	.8761	.0292	.8710	.8659	.9447	.9385
$HFCNet_{24}$ [24]	Hybrid	.8827	.0281	.8831	.8718	.9525	.9433
$MEANet_{24}$ [26]	MobileNetV2	.8661	.0325	.8625	.8587	.9399	.9284
$SAFINet_{24}$ [27]	MobileNetV2	.8718	.0338	.8628	.8577	.9375	.9332
$ISAANet_{24}$ [48]	MobileNetV3	.8732	.0281	.8760	.8696	.9497	.9431
$LGIPNet_{25}$ [64]	PVT	.8854	.0276	.8791	.8668	.9499	.9432
$MRBINet_{25}$ [65]	Res2Net	.8824	.0268	.8800	.8755	.9489	.9425
$BCARNet_{25}$ [66]	ResNet	.8757	.0306	.8689	.8651	.9407	.9365
$MambaDif_{26}$ [67]	VMamba	.8852	.0256	.8821	.8771	.9514	.9472
MSDEN$_{288\times288}$	MobileNetV2	.8702	.0319	.8665	.8612	.9409	.9337
†**MSDEN**$_{288\times288}$	MobileNetV2	.8735	.0299	.8689	.8629	.9439	.9376
†**MSDEN**$_{352\times352}$	MobileNetV3	.8754	.0309	.8713	.8659	.9444	.9385

† Adding extra multi-scale training strategy following [26], [27], and [48].

in the EORSSD and ORSSD dataset, which is increased by 0.0111 and 0.0150, respectively, than the best performance in NSI-SOD methods (PA-KRN). For computational cost, our method has 52.3× fewer parameters and 611.6× fewer FLOPs than PA-KRN. The superiority of our method suggests the importance of specific designs for ORSI targets.

Compared to conventional ORSI-SOD methods, our MSDEN achieves comparable performance while notably reducing computational costs. In some metrics, our lightweight model even outperforms others. For example, taking ACCoNet as an example, it achieves 0.8552 of the F_β^{mean} score in EORSSD, which is 0.0122 lower than MSDEN. Compared to the latest method, SFANet, MSDEN demonstrates comparable performance across metrics such as S_α, E_ξ^{mean}, and F_ξ^{mean} (0.9349 versus 0.9303, 0.9726 versus 0.9729, and 0.8680 versus 0.8674). MSDEN also achieves a parameter reduction of 9.3× and 7.6× in FLOPs compared to SFANet. The evaluation results on all three datasets also demonstrate that our method can achieve competitive performance to SFANet. Despite a notable performance gap between our MSDEN and the HFCNet, our approach achieves a 52.1× reduction in parameter size and 119.2× reduction in FLOPs compared to HFCNet. Although MambaDif achieves state-of-the-art performance by leveraging the strengths of Diffusion and Mamba, the high computational cost of the forward diffusion process hinders its ability to meet real-time computational requirements. The competitive balance between efficiency and accuracy in our approach validates the effectiveness of refining the intrinsic feature response of salient objects in the lightweight ORSI-SOD model.

2) Compared With Lightweight SOD Methods: As shown at the bottom of Table I, we compare our MSDEN with

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eight lightweight SOD methods for NSIs and ORSIs. We can observe that MSDEN achieves the best performance among previous lightweight NSI-SOD methods on EORSSD and ORSSD datasets. For lightweight ORSI-SOD, SeaNet [22], MEANet [26], and SAFINet [27] are specifically designed to reduce the number of parameters and FLOPS based on the MobileNetV2 encoder. Compared to these methods, our MSDEN exhibits a lower computational cost and a lower storage requirement. For example, MSDEN achieves 13% fewer parameters (2.7M versus 3.1M) and 86.7% lower computational cost (1.0G versus 7.6G FLOPs) compared to SAFINet, while delivering superior accuracy. In benchmark datasets, our method notably outperforms SeaNet in EORSSD and ORSSD datasets. For instance, MSDEN achieves higher S_α and F_β^{mean} (0.9303 versus 0.9208 and 0.8674 versus 0.8591) than SeaNet on the EORSSD dataset. For CorrNet and MEANet, MSDEN achieves superior performance on the EORSSD and ORSI-4199 datasets and comparable performance on the ORSSD dataset while exhibiting fewer parameters and FLOPs. MEANet, SAFINet, and ISAANet employ the multiscale training strategy in data augmentation. We use this training strategy to train our MSDEN and observe a significant performance improvement, particularly in the EORSSD and ORSI-4199 datasets. For example, the S_α metric improves from 0.9303 to 0.9341 on the EORSSD and from 0.8702 to 0.8735 on the ORSI-4199. These gains demonstrate that optimized training protocols can help bridge the accuracy gap between lightweight and computationally intensive ORSI-SOD architectures. The ISAANet adopts MobileNetV3 as its encoder and specifies an input image size of 352 $\times$ 352. For a fair comparison, we adopt MobileNetV3 as MSDEN's backbone while setting the input size to 352 $\times$ 352. The quantitative results in Table I demonstrate MSDEN's superior performance over ISAANet on the EORSSD and ORSSD benchmarks. For example, MSDEN achieves relative improvements of 0.0028 in S_α and 0.0015 in F_β^{mean} in the EORSSD. Compared with the recent lightweight architecture LiteSalNet, our method achieves superior performance on both EORSSD and ORSSD, while reducing FLOPs by 78%. On the challenging ORSI-4199 benchmark, our MSDEN outperforms ISAANet in S_α, while showing marginal decreases in other metrics. Although our model achieves a 77% reduction in FLOPs compared to ISAANet, its performance decline on the ORSI-4199 benchmark underscores the fundamental challenge of maintaining representation capacity within extremely efficient architectures. Furthermore, our experiments quantify the impact of performance on input resolution, showing that E_ξ^{mean} decreases from 0.9794 to 0.9764 when the input size is reduced from 352 $\times$ 352 to 288 $\times$ 288 on the ORSSD dataset. This finding has inspired us to develop an efficient ORSI-SOD architecture that maintains high resolution.

3) Computational Complexity Comparison: To further assess the computational complexity of our method relative to recent lightweight ORSI-SOD methods, we compare the FPS and parameter size of our method with those of SeaNet [22], CorrNet [25], MEANet [26], and SAFINet [27] as illustrated in Fig. 1. In Fig. 1, we observe that MSDEN outperforms existing lightweight ORSI-SOD models while maintaining a smaller parameter size. Moreover, we convert these lightweight ORSI-SOD models into ONNX and deploy them on a resource-constrained HUAWEI Mate 70 device to evaluate their efficiency. For fair comparison, the input size is set to 288 $\times$ 288. As illustrated in Fig. 1, while CorrNet achieves optimal performance, its FPS of only 1.3 limits its practical applicability. Although MEANet and SAFINet have achieved significant parameter reduction with higher accuracy, their inference speed is still not ideal (MEANet at 5.5 FPS and SAFINet at 5.0 FPS), limiting the potential for real-time deployment. SeaNet delivers a competitive inference speed of 9.9 FPS, yet its accuracy remains substantially inferior to leading state-of-the-art methods. By contrast, MSDEN attains nearly 2$\times$ acceleration over SeaNet while achieving superior performance, thereby highlighting its outstanding efficiency–accuracy trade-off.

4) Qualitative Comparison: We present the qualitative comparison of our MSDEN and ten state-of-the-art ORSI-SOD methods on four challenging ORSI scenes in Fig. 5. In the scene of large salient objects (a), we observe that lightweight methods such as SeaNet and CorrNet tend to discard specific salient regions, whereas other methods like PA-KRN are proficient in recognizing most objects, but face challenges in precisely localizing object boundaries and introducing background clutter. Our approach excels in object localization and boundary refinement, due to the local contrast learning of MDEGM and the global feature refinement of CSFM. For scene (b) featuring multiple small objects, we observe that previous methods like MEANet and CorrNet tend to emphasize some redundant targets and are challenging to detect densely packed small targets. In contrast, our MSDEN effectively suppresses background objects, facilitating precise discrimination of salient objects. For the salient objects with irregular typologies (c), our method can effectively capture the overall structure and produce smooth edges of the object. This capability is attributed to the dynamic interaction between local contrast and target semantic information in our approach. The scene involving salient objects with background interference (d) illustrates a common challenge in ORSIs. Taking the second column in (d) as an example, the low contrast between the drainage grate and the background poses a challenge for state-of-the-art methods such as MEANet and SFANet in effectively distinguishing between the target and the background. In contrast, methods such as HFCNet excel in object localization, but pose difficulties in capturing edge structures. Benefiting from MDEGM's perception of local structural boundaries, our approach is capable of not only locating salient objects but also revealing boundary structures. This highlights the advantage of our method for handling scenes with low contrast and background interference.

5) Challenging Example Comparison: As aforementioned, the suppression and interference caused by complex backgrounds hinder low-capacity models in extracting discriminative features to distinguish foreground pixels from background pixels, particularly in complex ORSI scenarios. We select some challenging examples in Fig. 6. When the salient objects closely resemble the complex background, lightweight ORSI-SOD models such as MEANet and SAFINet pose challenges in accurately identifying objects, resulting in numerous missed detections and false positives. In comparison, HFCNet, with its extensive parameter space, effectively extracts discriminative patterns and achieves notably better performance. We can

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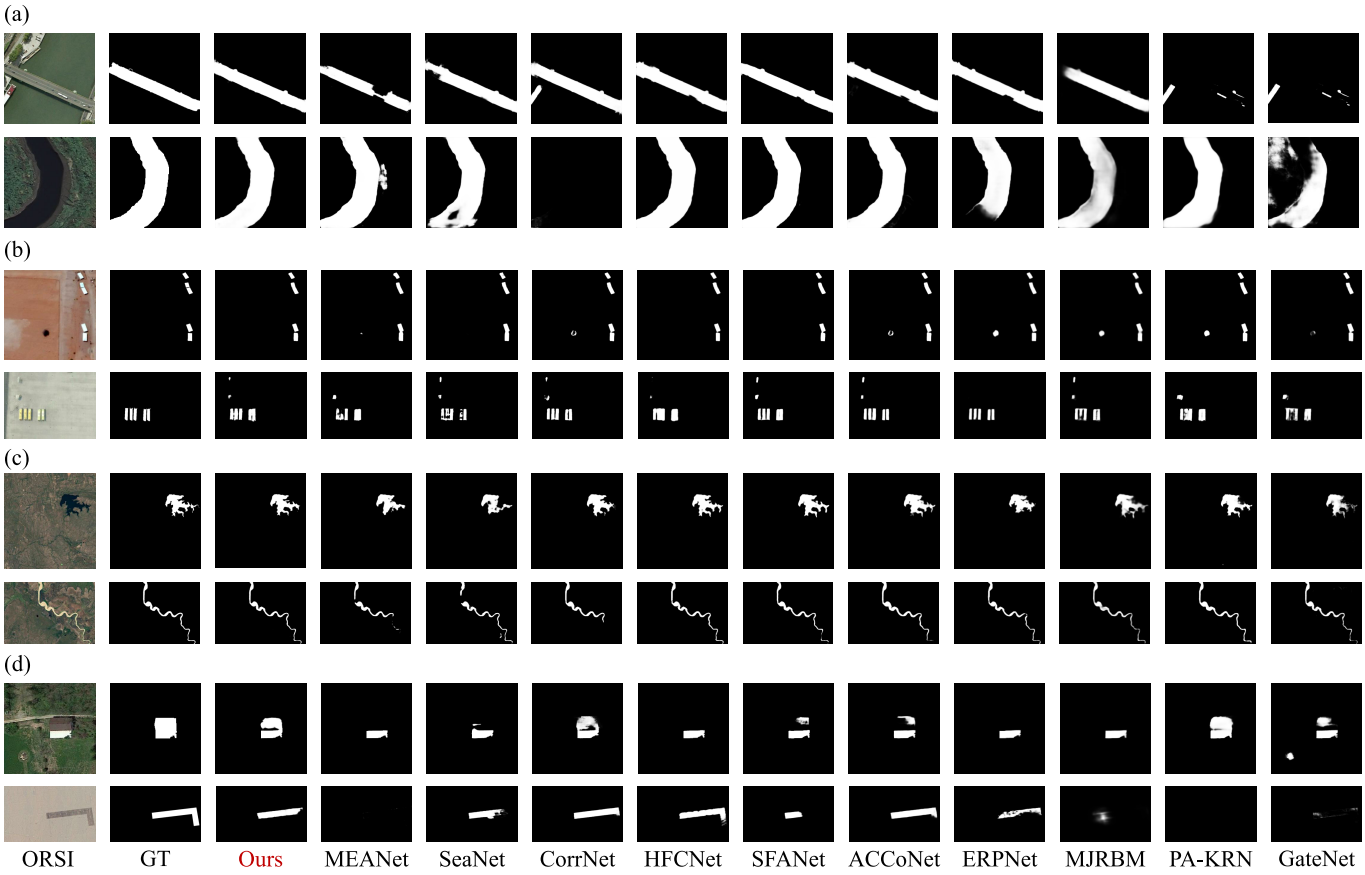


Fig. 5. Qualitative comparisons with ten representative state-of-the-art methods, including MEANet [26], SeaNet [22], CorrNet [25], HFCNet [24], SFANet [34], AccoNet [23], ERPNet [63], MJRBM [32], PA-KRN [62], and GateNet [60].

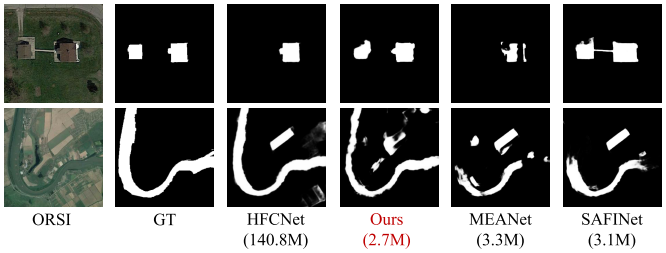


Fig. 6. Illustrative challenging examples. We compare the saliency maps predicted by our method against HFCNet [24], MEANet [26], and SAFINet [27].

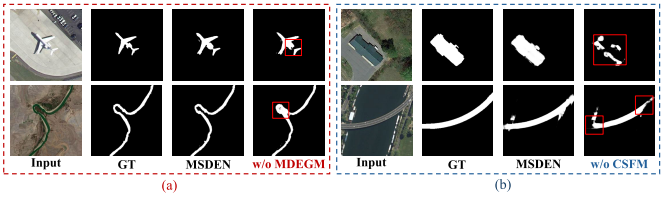


Fig. 7. Qualitative comparisons with (a) w/o MDEGM and (b) w/o CSFM.

observe that our method enables robust detection capability against cluttered backgrounds with low contrast, demonstrating the effectiveness of optimizing the boundary characteristics of salient objects.

C. Ablation Study on the MSDEN

1) Ablation Study on the Proposed Components: In this section, we evaluate the effectiveness of the proposed MDEGM and CSFM, respectively. As shown in Table III, we can observe that the proposed MDEGM and CSFM notably enhance the performance of detecting salient objects in ORSIs. In w/o MDEGM, the MSDEN exhibits a significant performance drop. The S_α decreases from 0.9384 to 0.9218. This notable drop suggests the effectiveness of MDEGM in enhancing object response by refining boundary representation and mitigating background interference. In w/o CSFM, we apply a Conv1 $\times$ 1 layer to reduce the channel dimension of the MDEGM output to 24. CSFM is designed to capture long-range contextual information. The S_α on ORSSD decreases by 0.0092 when CSFM is removed. This result highlights the importance of modeling long-range contextual information to adapt to the scale diversity of objects in ORSIs. Furthermore, we provide the qualitative comparison of MSDEN, w/o MDEGM, and w/o CSFM in Fig. 7. In Fig. 7(a), we observe

TABLE III
ABLATION STUDY ON PROPOSED COMPONENTS

Variant	EORSSD				ORSSD			
	$S_\alpha \uparrow$	$F_\beta^{mean} \uparrow$	$E_\xi^{mean} \uparrow$	$\mathcal{M} \downarrow$	$S_\alpha \uparrow$	$F_\beta^{mean} \uparrow$	$E_\xi^{mean} \uparrow$	$\mathcal{M} \downarrow$
MSDEN	.9303	.8674	.9729	.0069	.9389	.8920	.9749	.0089
w/o MDEGM	.9204	.8465	.9559	.0079	.9218	.8725	.9592	.0126
w/o CSFM	.9258	.8598	.9665	.0081	.9296	.8859	.9692	.0109

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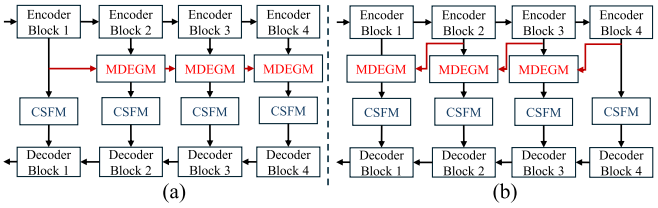


Fig. 8. Various designs of MSDEN. (a) MSDEN with a top-down pathway. (b) MSDEN with preguided MDEGM.

that w/o MDEGM results in a loss of structural detail, such as inaccurate detection of the aircraft's tail. Fig. 7(b) illustrates that w/o CSFM diminishes the detection performance for large-scale objects, leading to incomplete bridge detection.

2) Ablation Study on the General Architecture Design: Our MSDEN is built on a bottom-up pathway architecture, which uses the MDEGM output $\mathcal{G}^{i+1}$ to enhance the representation of the feature in the i−th stage (F^i). To demonstrate the efficiency of our design, we design two variants of MSDEN (*Top-down* and *Preguided*). As illustrated in Fig. 8, the specific details of these variants are presented below:

a) Top-down pathway: As illustrated in Fig. 8(a), we construct MSDEN using a top-down pathway, in contrast to the bottom-up approach. In this architecture, the shallow features $\mathcal{G}^i$ are progressively transmitted to deeper layers, guiding the learning of the deep semantic features in the $i+1$th stage;

b) Preguided: In contrast to using the output feature $\mathcal{G}^{i+1}$ of the MDEGM as the guidance in MSDEN, the preguided architecture employs the output features F^{i+1} in the i-th encoder stage to facilitate feature learning in the ith stage, as illustrated in Fig. 8(b);

The performance comparison of these variants can be observed in Table IV. In top-down architecture, shallow features often contain massive redundant information, which can interfere with the learning of high-level semantics, ultimately reducing performance. Table IV demonstrates the drawback of this architecture, with S_α decreasing from 0.9389 to 0.9292 in the ORSSD dataset. Moreover, the variation in guided features also affects performance. The accuracy of the preguided architecture is comparable to that of our method. However, the output of the feature by the encoder contains redundant information that can interfere with guidance, resulting in a 0.0067 decrease of S_α in the ORSSD dataset. In contrast, our method utilizes the postguided approach in the MDEGM to highlight crucial features and provide enhanced guidance for the features from the preceding layer, ultimately resulting in superior performance.

D. Ablation Study on the MDEGM Design

In this section, we conduct extensive ablation experiments on MDEGM design to further explore the feature guidance for the ORSI-SOD task.

1) Design of MSDEC: Compared to natural objects, SOD in ORSIs is based on local information with a limited receptive field. To further investigate the designed MSDEC, we employ multiple groups of different convolutional operators to extract rich boundary information. To ensure that no additional parameters or FLOPs are introduced, we divide the input features into N groups according to the channel dimension,

TABLE IV
ABLATION STUDY ON ARCHITECTURE DESIGN

Variant	EORSSD				ORSSD			
	$S_\alpha\uparrow$	$F_\beta^{mean}\uparrow$	$E_\xi^{mean}\uparrow$	$\mathcal{M}\downarrow$	$S_\alpha\uparrow$	$F_\beta^{mean}\uparrow$	$E_\xi^{mean}\uparrow$	$\mathcal{M}\downarrow$
MSDEN	.9303	.8674	.9729	.0069	.9389	.8920	.9749	.0089
Top-down	.9243	.8592	.9667	.0075	.9292	.8838	.9740	.0108
Pre-guided	.9270	.8601	.9672	.0071	.9322	.8871	.9741	.0102

TABLE V
ABLATION STUDY ON THE DILATION SETTINGS IN MSDEC

Dilation Setting	EORSSD				ORSSD			
	$S_\alpha\uparrow$	$F_\beta^{mean}\uparrow$	$E_\xi^{mean}\uparrow$	$\mathcal{M}\downarrow$	$S_\alpha\uparrow$	$F_\beta^{mean}\uparrow$	$E_\xi^{mean}\uparrow$	$\mathcal{M}\downarrow$
N=1	.9287	.8691	.9704	.0074	.9347	.8928	.9737	.0107
N=2	.9303	.8674	.9729	.0069	.9389	.8920	.9749	.0089
N=4	.9277	.8604	.9683	.0071	.9348	.8915	.9743	.0100

TABLE VI
ABLATION STUDY ON THE DIFFERENCE OPERATOR

Variants	EORSSD				ORSSD			
	$S_\alpha\uparrow$	$F_\beta^{mean}\uparrow$	$E_\xi^{mean}\uparrow$	$\mathcal{M}\downarrow$	$S_\alpha\uparrow$	$F_\beta^{mean}\uparrow$	$E_\xi^{mean}\uparrow$	$\mathcal{M}\downarrow$
Standard	.9279	.8619	.9681	.0071	.9342	.8919	.9742	.0097
CDC [39]	.9240	.8609	.9625	.0070	.9316	.8847	.9723	.0099
CPDC [40]	.9226	.8563	.9614	.0075	.9298	.8879	.9730	.0106
DEC	.9303	.8674	.9729	.0069	.9389	.8920	.9749	.0089

with each group assigned a convolution of a specific scale. In our experiments, we set N to 1, 2, and 4. The comparison results are presented in Table V. The different values of N yield comparable results. In particular, when $N = 2$, the model shows the best performance in terms of metrics S_α and $\mathcal{M}$, while slightly underperforming compared to when $N = \{1, 4\}$ in the E_ξ^{mean} metric. The experiment demonstrates that extracting multiscale boundary information in a grouped manner effectively enhances the perception of salient objects of varying scales. However, due to the lower number of channels in the lightweight model, setting N to 4 results in a limited representation of features on each scale, which consequently degrades the model performance.

2) Design of Kernel Variants: In this section, we provide the ablation study on selecting the convolutional operator. We conducted an ablation study by substituting the DEC with standard convolutional kernels, quantitatively evaluating the impact of gradient-aware convolution. Then we compare the performance of CDC [39], CPDC [40], and DEC. As shown in Table VI, we can observe that CDC and CPDC result in a performance drop, which demonstrates that the design of gradient-aware convolution is crucial for high-level tasks. CDC and CPDC compromise the model's representative capacity in pixel-level understanding tasks by overemphasizing edge information while failing to adequately capture semantic object features. As illustrated in Fig. 9, while CDC and CPDC enhance the delineation of boundaries, they also degrade saliency representation and introduce extraneous complex edge features from the background. Standard convolution inherently lacks the ability to extract edge information from salient objects, rendering it vulnerable to complex background interference. For example, in the case of ships, features within boundaries exhibit activation intensity close to the background, which ultimately degrades segmentation performance. In contrast, DEC better balances the enhancement of object

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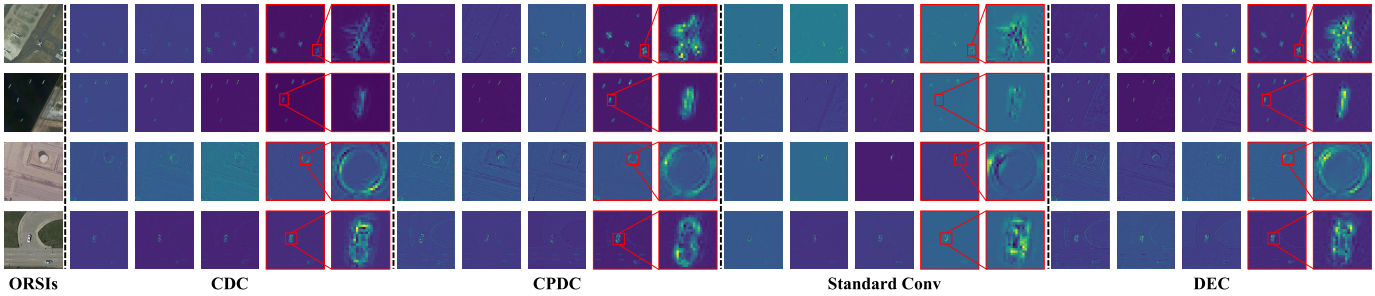


Fig. 9. Visualization of features generated by different kernel variants. The red box indicates the averaged features along the channel dimension.

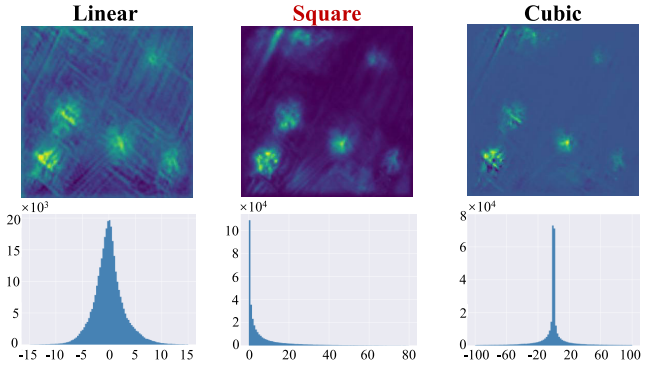


Fig. 10. Feature visualization (*the first row*) in cross-gate attention and the corresponding feature distribution histograms (*the second row*) under linear, square, and cubic settings.

TABLE VII

ABLATION STUDY ON THE POW OF FEATURE

pow	FLOPs	EORSSD			
		$S_\alpha \uparrow$	$F_\beta^{mean} \uparrow$	$E_\xi^{mean} \uparrow$	$\mathcal{M} \downarrow$
linear	0.53M	0.9276	0.8644	0.9680	0.0069
square	1.06M	0.9303	0.8674	0.9729	0.0069
cubic	1.59M	0.9194	0.8511	0.9608	0.0083

boundaries with the preservation of semantic consistency within objects.

3) High-Order Feature Analysis: In the CGA, we utilize the square term $\mathcal{G}_{edge}^{i+1} \odot \mathcal{G}_{edge}^{i+1}$ to model the semantic boundary indicator in the deep feature. To investigate the impact of feature order on capturing salient targets, we set the linear term $\mathcal{G}_{edge}^{i+1}$ and the cubic term $\mathcal{G}_{edge}^{i+1} \odot \mathcal{G}_{edge}^{i+1} \odot \mathcal{G}_{edge}^{i+1}$ as a comparative experiment. The result can be observed in Table VII. We can observe that the cubic term of $\mathcal{G}_{edge}^{i+1}$ exhibits significant performance degradation. The visualization can be found in Fig. 10. We observe that the utilization of square terms effectively suppresses background pixels, highlighting salient target regions, thus providing valuable guidance for the representation of shallow features. The pixel-wise square operation is a nonlinear point transformation that enhances signal energy or power by amplifying intensity deviations from the local mean. This mechanism inherently suppresses homogeneous background regions, which typically exhibit low intensity (approximately zero in normalized feature maps) and low variance. Moreover, it enhances contrast variations associated with foreground or target pixels that demonstrate higher intensities and variance. Such contrast amplification between foreground and background areas effectively emphasizes discriminative features, improving the representation of salient objects in

TABLE VIII

ABLATION STUDY ON EACH COMPONENT OF MDEGM

MDEGM			EORSSD				ORSSD			
MSDEC	CGA	BCA	$S_\alpha \uparrow$	$F_\beta^{mean} \uparrow$	$E_\xi^{mean} \uparrow$	$\mathcal{M} \downarrow$	$S_\alpha \uparrow$	$F_\beta^{mean} \uparrow$	$E_\xi^{mean} \uparrow$	$\mathcal{M} \downarrow$
			.9204	.8465	.9559	.0079	.9218	.8725	.9592	.0126
✓			.9236	.8517	.9596	.0073	.9274	.8844	.9653	.0113
	✓		.9242	.8566	.9585	.0078	.9267	.8797	.9642	.0110
		✓	.9227	.8520	.9560	.0084	.9233	.8721	.9602	.0119
✓	✓		.9272	.8610	.9651	.0072	.9300	.8867	.9703	.0111
✓		✓	.9267	.8598	.9661	.0074	.9281	.8791	.9684	.0109
	✓	✓	.9233	.8575	.9622	.0079	.9295	.8857	.9675	.0098
✓	✓	✓	.9303	.8674	.9729	.0069	.9389	.8920	.9749	.0089

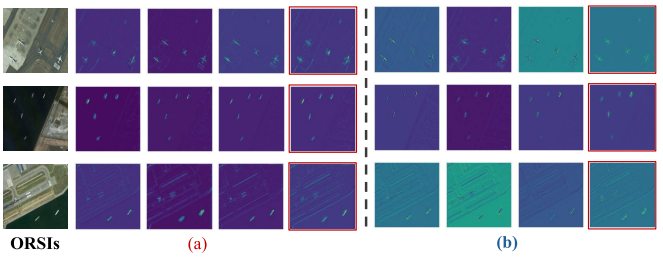


Fig. 11. Output feature maps in the first stage of MSDEN. The red box indicates the averaged features along the channel dimension. (a) Feature from MDEGM with MSDEC. (b) Feature from MDEGM with a standard convolutional filter.

local remote sensing image patches.. The linear term contains a significant amount of redundant background information, while the cubic terms suppress feature representation, leading to a negative impact on performance.

4) Effectiveness of Each Component of MDEGM: We separately analyze the impact of each component in MDEGM. The result can be observed in Table VIII. In w/o CGA, we replace the dot product fusion operation with direct addition. As shown in Table VIII, the accuracy decreases without the MSDEC. Taking S_α as an example, the accuracy drops from 0.9303 to 0.9233, demonstrating the significance of capturing local contrast information for the ORSI-SOD task. Furthermore, the observed performance degradation in the w/o CGA and w/o BCA highlights the effectiveness of the deep semantic feature guidance as well as the channel attention guided by local contrast information. Furthermore, we compare the impact of each individual component on model performance. The CGA and MSDEC notably improve the performance of the lightweight ORSI-SOD model, validating the importance of interlayer interactions and boundary enhancement.

5) Intermediate Feature Map Visualization: To further evaluate the effectiveness of our design, we visualize the intermediate feature maps of MDEGM in the first stage. As shown in Fig. 11, we provide the output of the MDEGM along

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TABLE IX
ABLATION STUDY ON CSFM

Model	Params	FLOPs	EORSSD			
			$S_\alpha \uparrow$	$F_\beta^{mean} \uparrow$	$E_\xi^{mean} \uparrow$	$\mathcal{M} \downarrow$
MSDEN	2.67M	1.01G	0.9303	0.8674	0.9729	0.0069
w/o CSFM	2.65M	0.92G	0.9258	0.8598	0.9665	0.0081
w/o CFAM	2.65M	0.92G	0.9274	0.8628	0.9653	0.0076
w/o CSAM	2.67M	1.01G	0.9292	0.8659	0.9704	0.0077

TABLE X
ABLATION STUDY ON CHANNEL CONFIGURATION IN CSFM

Channel	Params	FLOPs	ORSSD			
			$S_\alpha \uparrow$	$F_\beta^{mean} \uparrow$	$E_\xi^{mean} \uparrow$	$\mathcal{M} \downarrow$
12	2.7M	1.0G	0.9400	0.8980	0.9733	0.0091
24	2.7M	1.3G	0.9415	0.8971	0.9765	0.0090
36	2.8M	1.5G	0.9411	0.8956	0.9716	0.0089

TABLE XI
EVALUATION ON EORSSD W/O IMAGENET PRETRAINING

Methods	Backbone	$F_\beta^{max} \uparrow$	$F_\beta^{mean} \uparrow$	$E_\xi^{max} \uparrow$	$E_\xi^{mean} \uparrow$
ACCoNet	ResNet	0.8574	0.7988	0.9633	0.9459
CorrNet	L.W. VGG	0.8489	0.8307	0.9470	0.9097
SeaNet	MobileNet	0.8034	0.7902	0.9377	0.9299
MSDEN	MobileNet	0.8112	0.7940	0.9411	0.9333

with the results processed through MSDEC. Additionally, we compare the differences between MSDEC and the standard convolution kernel in MDEGM. Our observations indicate that the integration of local contrast cues in MSDEC highlights the pixel regions surrounding the boundary of salient objects while effectively suppressing redundant background information. This suggests that the enhancement of local contrast cues effectively refines the detailed information of salient objects and facilitates more reliable guidance of deep features.

E. Ablation Study on the CSFM

In this section, we provide the ablation study for CSFM. The CSFM aims to refine feature representation after MDEGM. We separately exclude the CFAM and CSAM modules from CSFM and evaluate the performance on EORSSD. The result can be observed in Table IX. We observe a decrease in performance when the feature is not refined by CSFM. w/o CFAM restricts the model to local receptive fields, limiting its ability to capture relationships between pixels over long distances, resulting in a decrease of 0.0046 in the F_β^{mean} metric. CSAM is utilized to amplify the importance of salient object regions in the spatial dimension. Its exclusion results in a decrease of 0.0015 in the F_β^{mean} metric.

In CSFM, the channel number is set to 12 to maintain a balance between model efficiency and representational capacity. To validate this configuration, we perform ablation experiments using 24 and 36 channels. As shown in Table X, increasing the number of channels yields only marginal improvements in both S_α and $\mathcal{M}$ metrics, while increasing computational overhead and parameter count. We therefore select 12 channels as the final design to achieve an optimal trade-off between accuracy and efficiency.

F. Comparison on w/o Pretrained Encoder

Contemporary ORSI-SOD methods rely on encoders pretrained on ImageNet. In this section, we evaluated the performance of different methods trained from scratch. As depicted in Table XI, we compared the performance of the methods with various encoders, including ACCoNet (based on ResNet), CorrNet (based on VGG), SeaNet, and our MSDEN (based on MobileNetV2). We observed that when using lightweight encoders, omitting pretrained weights leads to significant accuracy degradation. For example, CorrNet experienced a decrease of 0.0289 in the F_β^{max} metric, while SeaNet and our method exhibited reductions of 0.0615 and 0.0705, respectively. This indicates that the representational capacity of lightweight models in a limited parameter space relies more heavily on the pretrained parameters. When trained from scratch, our method also outperforms SeaNet in various metrics, such as F_β^{max} and F_β^{mean} (0.8112 versus 0.8034 and 0.7940 versus 0.7902).

TABLE XII
TEN TIMES RUNNING RESULTS OF LIGHTWEIGHT ORSI-SOD MODELS ON EORSSD

Method	FPS ↑	$S_\alpha \uparrow$	$F_\beta^{mean} \uparrow$	$E_\xi^{mean} \uparrow$	$\mathcal{M} \downarrow$
SeaNet	9.9	0.9184±0.0008	0.8472±0.0007	0.9607±0.0020	0.0081±0.0003
MEANet	5.5	0.9281±0.0017	0.8627±0.0038	0.9664±0.0020	0.0074±0.0005
SAFINet	5.0	0.9278±0.0011	0.8689±0.0056	0.9663±0.0023	0.0076±0.0010
MSDEN	15.9	0.9290±0.0011	0.8634±0.0028	0.9693±0.0029	0.0070±0.0001

G. Effectiveness of the Proposed Method.

To further validate the effectiveness of the proposed method, we retrained SeaNet, MEANet, and SAFINet 10 times on the EORSSD dataset, following SeaNet's training strategy. To ensure a fair comparison, all four models were implemented with a MobileNetV2 backbone. The mean values and standard deviations of the four evaluation metrics are summarized in Table XII. The results demonstrate that our approach outperforms existing lightweight ORSI-SOD methods.

V. LIMITATION AND FUTURE WORK

Despite the notable advancements achieved by MSDEN in the lightweight ORSI-SOD task, several limitations persist: 1) a considerable performance gap remains compared to large-parameter models, such as HFCNet; 2) due to the limitations of existing datasets, there is a lack of further exploration and validation on large-scale datasets; and 3) robustness to real-world deployment requires further explored including imaging degradation and environmental disturbances.

To further enhance the practical applicability of ORSI-SOD, we identify four key future research directions, including:

A. Large-Scale Training Dataset

Existing ORSI-SOD datasets suffer from limited scale and scene diversity, which constrains the generalization of the model in real-world applications. Therefore, it is crucial to collect a large-scale, diversified benchmark dataset that covers various geographical scenarios and target types. Furthermore, exploring unsupervised/weakly supervised learning for ORSI-SOD can alleviate the heavy reliance on manual annotations, thereby addressing the inherent annotation challenges in ORSIs.

B. Foundation Model Knowledge Driven

The existing ORSI-SOD methods depend on pretraining knowledge from ImageNet to alleviate the performance degradation caused by the limited training dataset. Empowering

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ORSI-SOD with knowledge of the vision foundation model trained on large-scale natural images, such as SAM, DINO, etc., is important for future research. In addition, comparing the performance gains of remote sensing and generic vision foundation models on the ORSI-SOD task is also worth further exploration.

C. Domain-Specific Architecture Design

The existing ORSI-SOD frameworks lack exploration of the characteristics of remote sensing scenes, such as sparsity. In future work, we will attempt to further design an ORSI-SOD framework guided by scene prior knowledge. By incorporating specific designs to extract key features from the scene, the performance gap between lightweight and large models can be further narrowed.

D. Real-World Robustness

Current ORSI-SOD methods exhibit notable vulnerability to real-world image degradation, including sensor noise, atmospheric scattering, and weather-induced distortions. Therefore, we will investigate the noise-invariant representation and the test-time adaptation method for ORSI-SOD, which facilitates the maintenance of model performance under varying weather conditions and imaging imperfections.

VI. Conclusion

In this article, we propose a lightweight SOD architecture specifically designed for ORSIs, referred to as MSDEN. The MSDEN combines local contrast feature modeling and long-range contextual information capture in a unified encoder–decoder structure. To refine the intrinsic feature response of objects suppressed by complex ORSIs' backgrounds, we introduce the MDEGM, which leverages prior visual cues to enhance the representation of salient object structures. Specifically, MDEGM captures multiscale local semantic discrepancies across neighboring layers and highlights salient object features through deep feature guidance. Subsequently, the CSFM enhances the features by leveraging long-range contextual relationships to refine the representation of objects with diverse scales. Finally, a lightweight decoder progressively aggregates features from deeper layers to shallower layers, facilitating the generation of accurate saliency maps. Incorporating these specific designs, our MSDEN achieves performance comparable to recent state-of-the-art methods while maintaining high efficiency.

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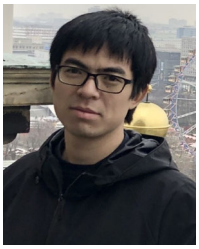
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